Replication of Sampling to Learn Words by Zettersten & Saffran (2020, Developmental Science)

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Introduction

Proposal:

Zettersten & Saffran (2020) uses cross situational learning tasks to understand how learners gather new information for word learning. What do people choose to learn when presented with an uncertainty of new words? Experiment 1 asked whether or not adult participants,

when given information to sample from, would choose information that reduces referential ambiguity. What sampling strategies are people using for word learning? This is the central question that they later ask with a developmental population. They find that people choose the ambiguous word-object pair 62% of the time. The authors also ran a second version of Experiment 1 (S1), where the word-object ambiguity was graded instead across two between-subject conditions (Fully Ambiguous vs Partially Ambiguous). They replicated the findings from Experiment 1 and additionally also found that adults sampling preferences were sensitive to graded ambiguity.

This study aims to replicate the findings from the Supplementary Experiment 1 (S1) and also extend the study to see how sampling behavior changes over time by increasing the number of training-sampling rounds. We propose to first have participants complete S1 that is identical in structure and stimuli to replicate the original findings after which we would add the additional training-sampling rounds.

Methods

Power Analysis

Original effect size, power analysis for samples to achieve 80%, 90%, 95% power to detect that effect size. Considerations of feasibility for selecting planned sample size.

Planned Sample

Participants will be undergraduate students from UC San Diego recruited online via the SONA systems. Participants will be given course credits for participation. Exclusion criteria include 1) if they do not pass the auditory attention check, which includes listening to two recordings of a familiar English word ("apple" and "ball") and typing out the word into a free response text box 2) Only native English speakers will be able to participate in this study. Participants will be randomly assigned either the Fully Ambiguous condition or Partially Ambiguous condition.

Materials

The stimuli will be identical to those used in the original study, which consists of the following – "The objects were eight images of novel 'alien' creatures used in previous studies (Partridge et al., 2015). Eight novel words (beppo, finna, guffi, kita, noopy, manu, sibu, tesser) were recorded by a female native speaker of English and normalized in duration and average loudness. The association between each label and its target referent and the roles of the stimuli within a condition were randomized across participants. The stimuli were presented using jsPsych (de Leeuw, 2015). All stimuli and experimental scripts for Experiment 1 and all subsequent

experiments can be viewed on the Open Science Framework (OSF) (https://osf.io/udmvh/)." For this project, the experiment will be re-implemented using the latest version of jsPsych 8.

Procedure

The web experiment will be hosted on Github Pages and data will be collected to a Google Drive using the DataPipe plugin. The URL for the experiment will be available once the participant signs up for the study on SONA.

The design of this experiment will closely follow the original study which consisted of the following phases:

Training Phase "Participants completed 24 cross-situational learning trials (two blocks of 12 trials), presented in random order. Participants were instructed that their goal was to learn the association between eight novel labels and their referents. On each trial, participants were presented with two referents and two labels. The labels appeared sequentially in random order, both visually and auditorily. Consequently, the association between a particular label and its referent remained ambiguous on any single trial, but could be disambiguated by aggregating information across trials. Each object and its label occurred six times across the 24 training trials." Participants will be randomly assigned one of two conditions: Fully Ambiguous or Partially Ambiguous. In the Fully Ambiguous condition "half of the object-label pairs were never disambiguated and half of the object-label pairs were disambiguated across training trials". For an object-label pair to never be disambiguated means that it always co-occurred, so the learners are not able to work out which object goes with which label. In the Partially Ambiguous condition, "half of the object-label pairs were grouped such that two specific objects co-occurred on 4 out of their 6 occurrences, but each object occurred with a different object from the ambiguous object set on the remaining 2 trials The other four objects were disambiguated as in the Fully Ambiguous condition." Participants saw each of the 8 individual object-label pairs equally frequently.

Sampling Phase "Participants completed four sampling trials. On each trial, all eight objects appeared in randomized locations. Participants were instructed to select which of the eight items they wanted to hear on the next cross-situational learning trial. After participants made a selection, a second object was chosen at random from the remaining objects. The two objects and their labels then appeared together in a cross-situational word learning trial with the same structure as in the training phase."

Test Phase To assess whether or not participants learned the object-label associations , participants were asked to complete an 8-AFC (alternative forced choice) recognition test. "On each test trial, all eight objects appeared in randomized locations on the screen, along with one of the eight labels. Participants were asked to select the object that went with the label. No feedback was provided. Participants were tested on each label in random order, for a total of eight test trials."

Participants will complete the phases in the order Training-Test-Sampling-Test as in Experiment S1. Additionally, to understand how people's sampling behavior changes over time, participants will then complete two further Sampling-Test rounds, each with a Sampling Phase with 4 trials and Testing Phase with 8 trials using the same eight object-pair labels.

Analysis Plan

Data Pre-processing We will exclude observations from our analyses if participants fail to get the auditory attention check correct. Since participants' responses were free text, responses will be evaluated with a case-insensitive regex pattern.

We will assess two key dependent measures:

1. Sampling choices, specifically whether or not they choose ambiguous object-label pairs to learn from. We will fit the same logistic mixed-effects model used in Experiment S1, specified as follows:

```
ambiguous selection ~ 1 + condition + (1 | participant) + (1 | item)
```

2. Comparison of test accuracy before and after sampling. We will fit the same logistic mixed-effects model used in Experiment S1, specified as follows:

```
correct choice  test block × item type × condition +
(1 + test block × item type | participant) + (1 | item)
```

Differences from Original Study

1. We will be adding two additional Sampling-Test cycles, adding four additional blocks to the original experiment.
2. In the orignal study, Experiment S1 was conducted in-person and participants for Experiment 1 were recruited via MTurk. We will be recruiting participants from an undegraduate SONA pool and is run online.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or "none".

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.